

Lab Assignment #03

Aim:

Design, simulate, and analyze E-plane and H-plane Tee junctions using the rectangular waveguide from Experiment 01, and investigate the effects of design parameter variations on their scattering parameters.

Theory:

A waveguide Tee junction is a three-port network used for power division or power combining, formed by intersecting a secondary waveguide side arm with a main waveguide. The behavior of the junction is fundamentally determined by whether the side arm is oriented to couple with the Electric (E) field or the Magnetic (H) field of the dominant mode.

E-plane Tee

The E-plane (Series) Tee is constructed by attaching the side arm to the broader wall of the main waveguide so that the axis of the side arm remains parallel to the E-field of the TE₁₀ mode. In this configuration, when power enters through the side arm (Port 3), the E-field lines split into the main arm with opposite polarities, resulting in a 180 degree phase difference between the two outputs at Ports 1 and 2. For an ideal, lossless, and matched E-plane Tee, the scattering matrix (S-matrix) is defined by $S_{13} = -S_{23}$, representing its role as a phase-inverted power divider or a subtractor.

$$[S] = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{2} & \frac{1}{2} & -\frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

Fig 1.1 S matrix of E-plane Tee Waveguide

H-plane Tee

Conversely, the H-plane (Shunt) Tee is formed by attaching the side arm to the narrower wall of the main waveguide, keeping the side arm axis parallel to the H-field. When power is fed into the side arm, it splits symmetrically into the main arm with the E-field remaining in the same direction for both output ports. This produces signals that are perfectly in phase, characterized by $S_{13} = S_{23}$ in the scattering matrix. This junction acts as a coherent power adder or an in-phase power divider.

$$[S] = \begin{bmatrix} \frac{1}{2} & -\frac{1}{2} & \frac{1}{\sqrt{2}} \\ -\frac{1}{2} & \frac{1}{2} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} & 0 \end{bmatrix}$$

Fig 1.2 S matrix of H-plane Tee Waveguide

Parametric Sweep and Modal Analysis

The performance of these Tee junctions is heavily dependent on the wider dimension (a) of the waveguide. By performing a parametric sweep on using values of 25 mm, 34.85 mm, and 45 mm, we investigate how physical geometry alters wave propagation. The cutoff frequency (f_c) for the TE₁₀ mode is governed by the relationship $f_c = c/2a$. When a is reduced to 25 mm, the cutoff frequency increases to approximately 6 GHz. This shift can cause the waveguide to enter an evanescent state if the operating frequency is lower, leading to significant signal attenuation. At the standard 34.85 mm, the f_c is 4.3 GHz, providing an optimized margin for the C-band. Increasing the width to 45 mm lowers the f_c to 3.33 GHz, which, while allowing easier propagation, increases the risk of exciting higher-order modes like TE₂₀ that can degrade signal purity.

Furthermore, the characteristic impedance (Z_0) and guide wavelength (λ_g) are functions of the dimension a . At the standard 34.85 mm width, the junction is physically optimized for a specific impedance, resulting in minimized return loss (S_{11} and S_{33}). Deviating to 25 mm or 45 mm creates an impedance mismatch that increases reflections and elevates the return loss in dB. Simultaneously, changes in the guide wavelength alter the electrical length of the junction arms, causing phase shifts in the scattering parameters that are critical for applications requiring precise phase control between the output ports.

Table 1 - Parametric Details of WR-137

Parameter	Value
Broad Dimension (Width)	34.8488 mm
Narrow Dimension (Height)	15.7988 mm
Cutoff Frequency of Lowest Order Mode:	4.301 GHz
Cutoff Frequency of Upper Mode:	8.603 GHz
Recommended Frequency Band:	5.85 to 8.20 GHz
Length	100 mm (suggested)
Wall Material	PEC
Core Material	Air

Procedure:

1. Open CST Studio Suite and select the New Template option from the main dashboard.
2. Navigate to the Microwave & RF / Optical folder, choose Waveguides & Components, and then select Waveguides.
3. Choose the Time Domain Solver from the available options to handle the broadband frequency sweep.
4. Set the project units by defining Dimensions in mm and Frequency in GHz.
5. Set the simulation range with a Minimum of 2 GHz and a Maximum of 10 GHz to observe the fundamental TE₁₀ mode and higher-order mode transitions.
6. Open the Modeling tab and set the Background Material to PEC (Perfect Electric Conductor) with zero spacing in all directions.
7. Create parameters in the Parameter List for the wide dimension ($a=34.8488$), the narrow dimension ($b=15.7988$), and the length ($l=100$).
8. Use the Brick tool to create the main waveguide arm with dimensions based on parameters a , b , and l , assigning the material as Air.
9. Create a second Brick for the side arm, orienting it on the broad wall for an E-plane Tee or the narrow wall for an H-plane Tee.
10. Merge the two bricks using the Boolean Add operation to create a single, continuous three-port junction.
11. Use the Pick Face tool to select the openings of the three arms and define them as Waveguide Ports 1, 2, and 3.
12. Navigate to the Simulation tab and define Field Monitors for the Electric (E) and Magnetic (H) fields at 6 GHz.
13. Open the Parametric Sweep tool, add the parameter a as the sweep variable, and define the values at 25 mm, 34.8488 mm, and 45 mm.
14. Start the Setup Solver to run the simulation across all defined parametric steps.
15. Navigate to the 1D Results folder to view S-parameter plots and the 2D/3D Results folder to visualize the field distributions at the monitored 6 GHz frequency.

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

3D Model:

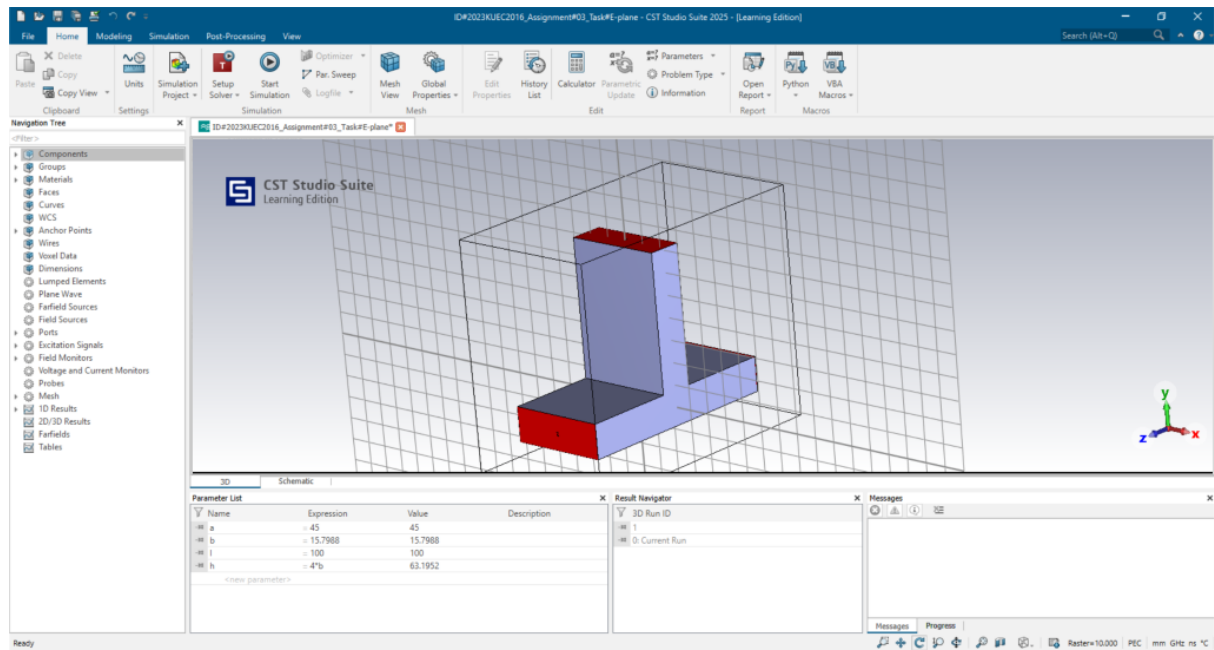


Fig 2.1 E Plane Tee Junction 3D Model

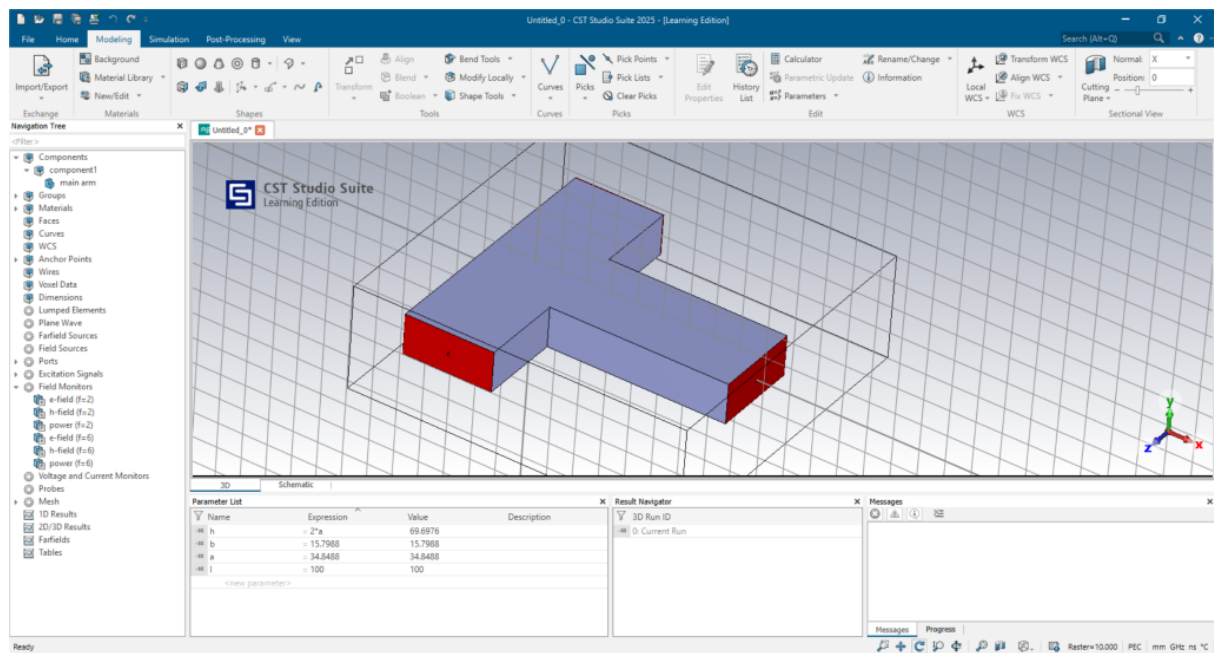


Fig 2.2 H Plane Tee Junction 3D Model

Results:

The electromagnetic characteristics of the WR-137 waveguide were simulated over a broadband frequency sweep from **2 GHz to 10 GHz**. This wide bandwidth allowed for the observation of the fundamental TE₁₀ mode as well as the "turn-on" points for higher-order modes (TE₂₀, TE₀₁, TE₁₁, etc.). Field distributions were captured at 3, 10, and 15 GHz to visualize the physical transition from evanescent to propagating states.

E Plane Tee Junction

Scattering Parameters

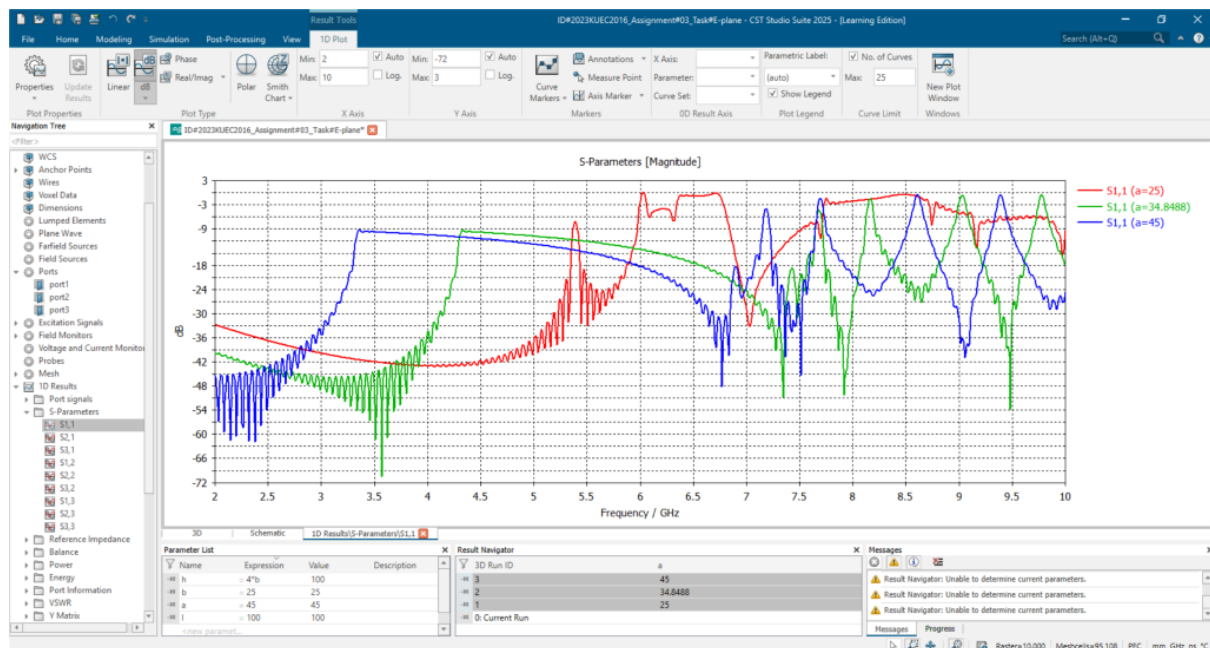


Fig 3.1 Scattering Parameter S(1,1) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

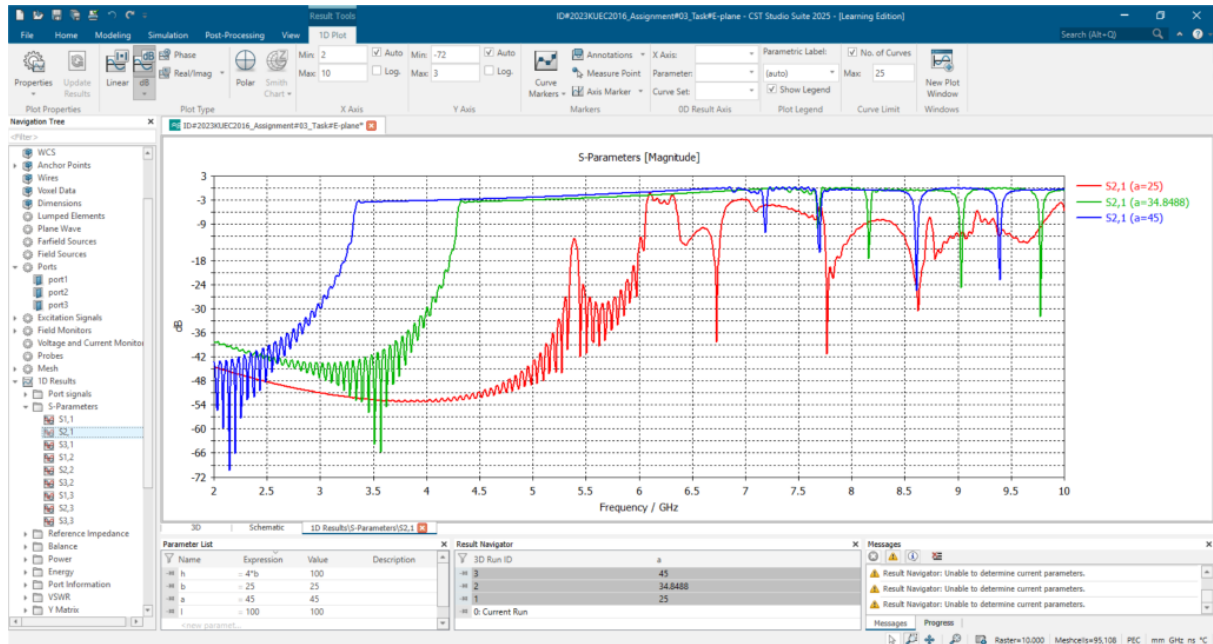


Fig 3.2 Scattering Parameter $S(2,1)$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

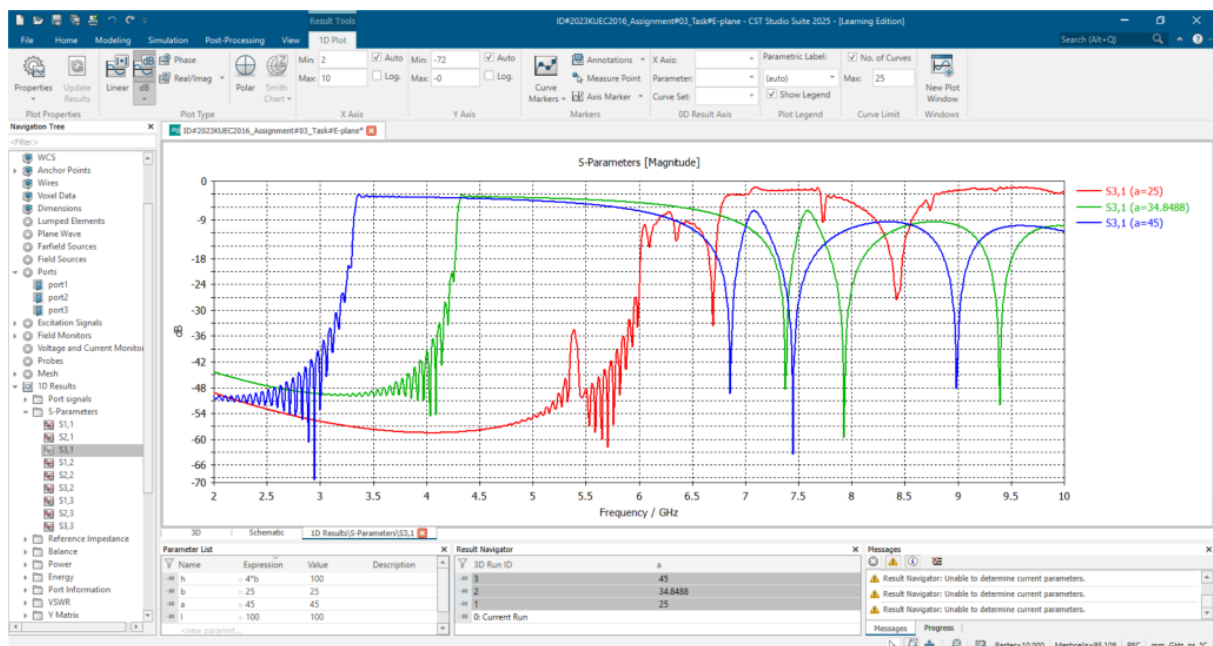


Fig 3.3 Scattering Parameter $S(3,1)$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

Submitted by
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Roll Number.: 2023KUEC2016

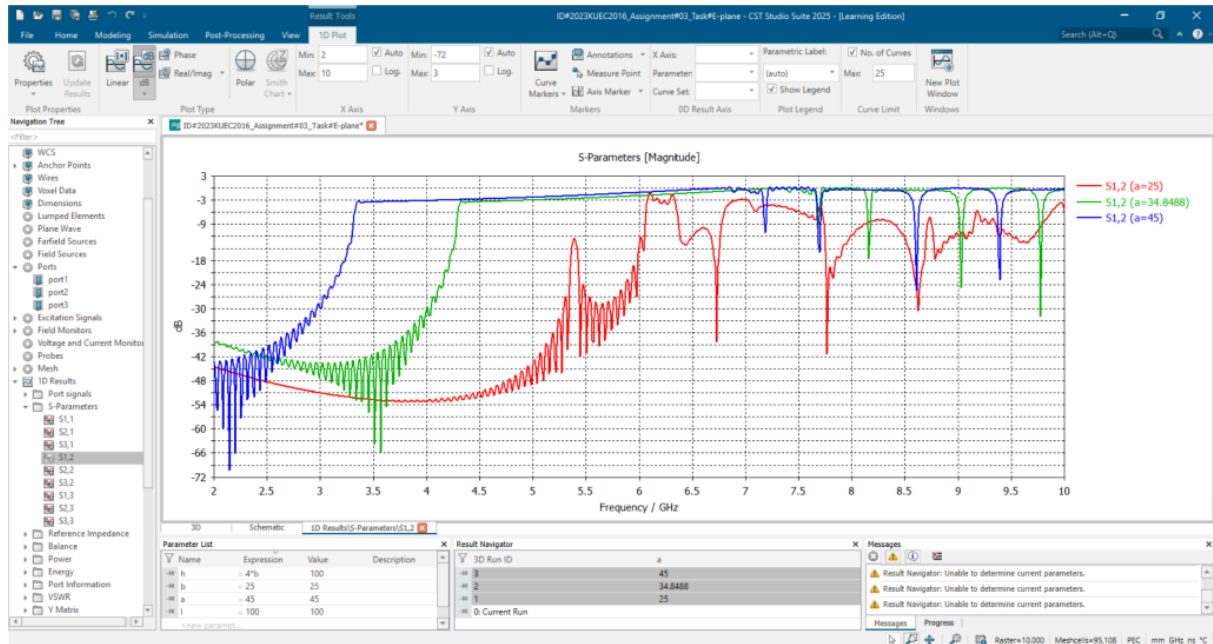


Fig 3.4 Scattering Parameter $S_{1,2}$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

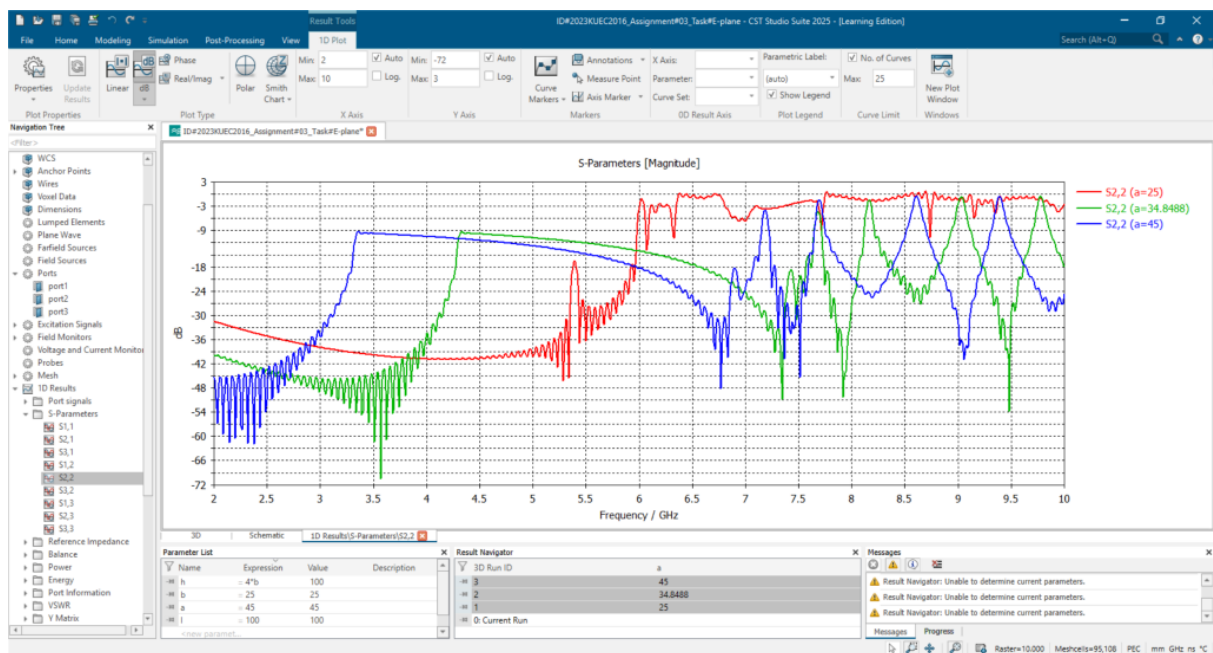


Fig3 .5 Scattering Parameter $S_{2,2}$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

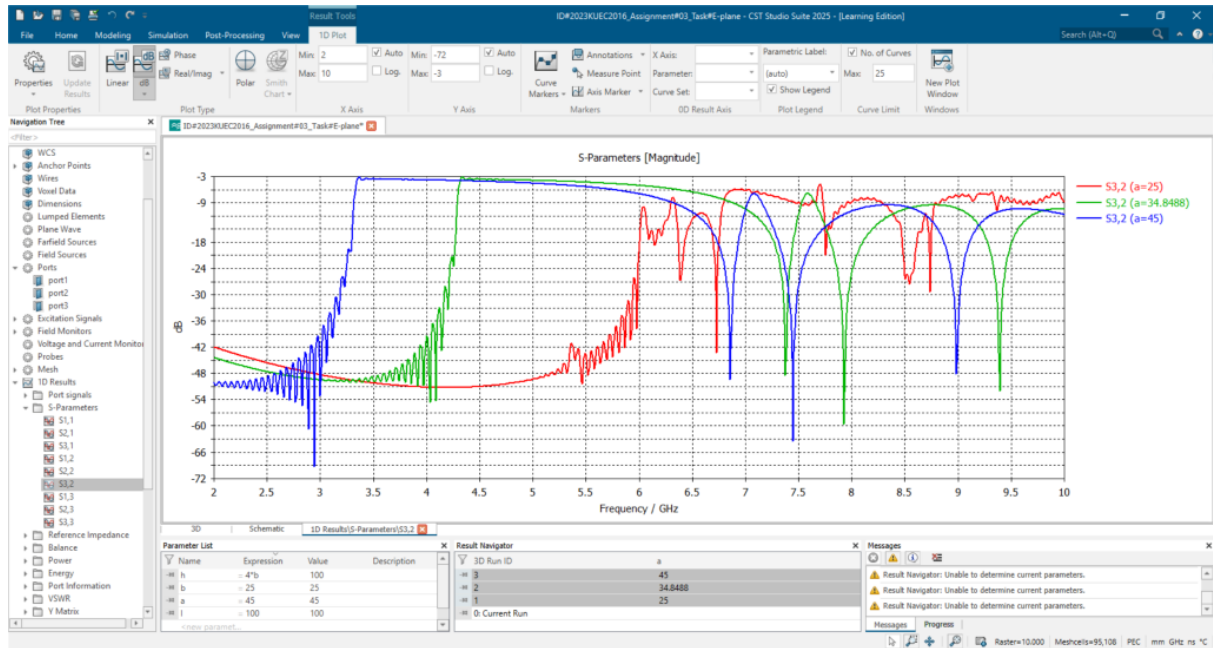


Fig 3.6 Scattering Parameter $S(3,2)$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

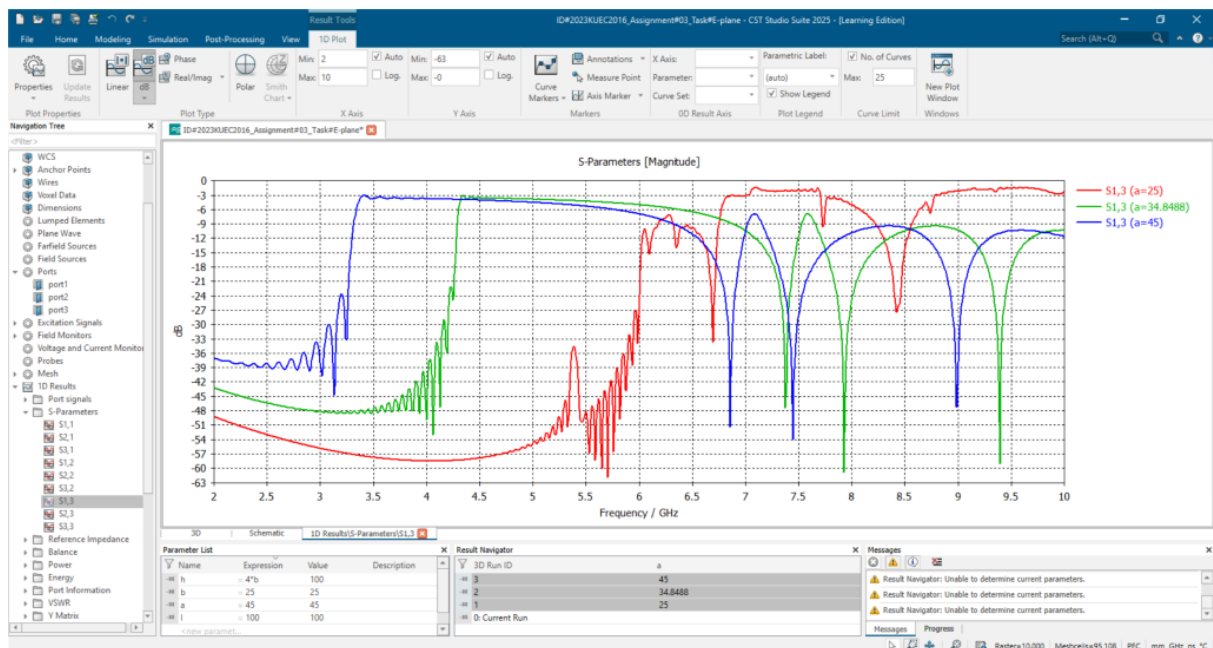


Fig 3.7 Scattering Parameter $S(1,3)$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

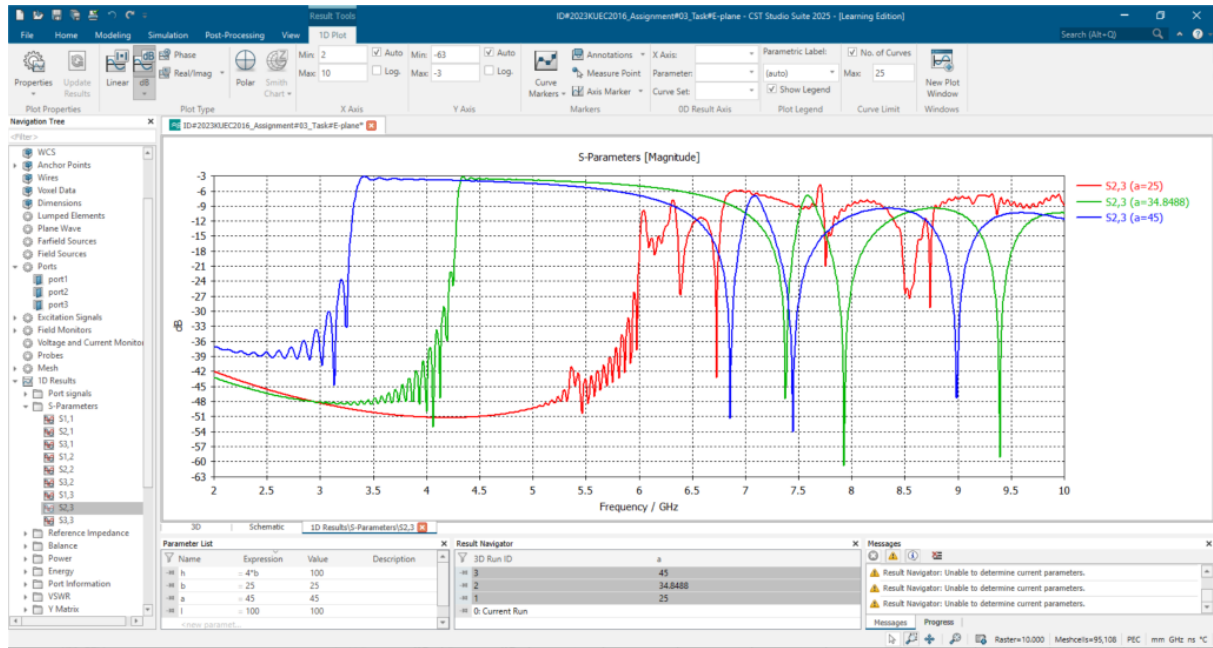


Fig 3.8 Scattering Parameter S(2,3) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

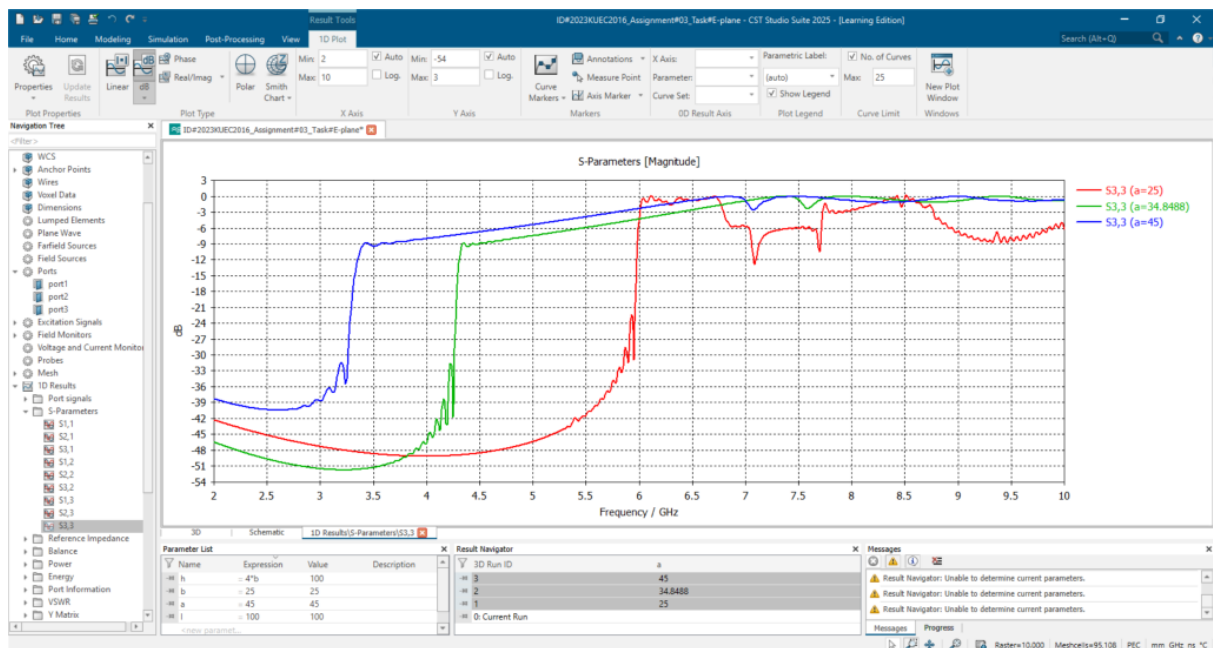


Fig 3.9 Scattering Parameter S(3,3) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

Electric Field Distributions

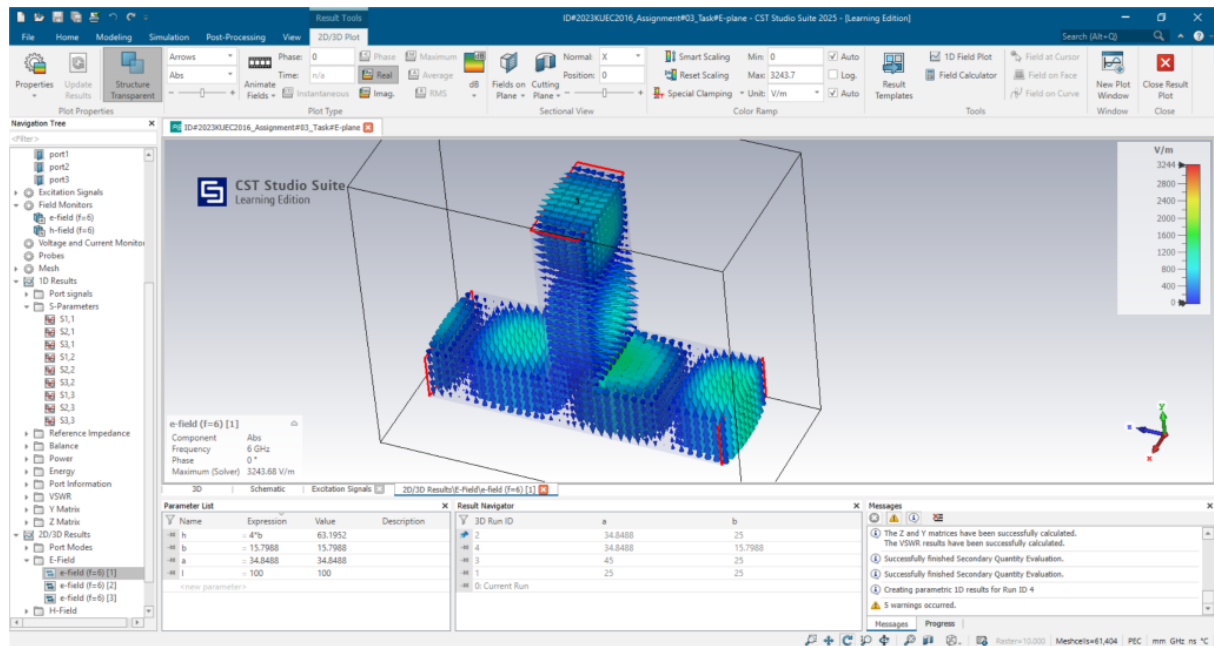


Fig 4.1 Electric field distribution for $f = 6$ GHz at port 1

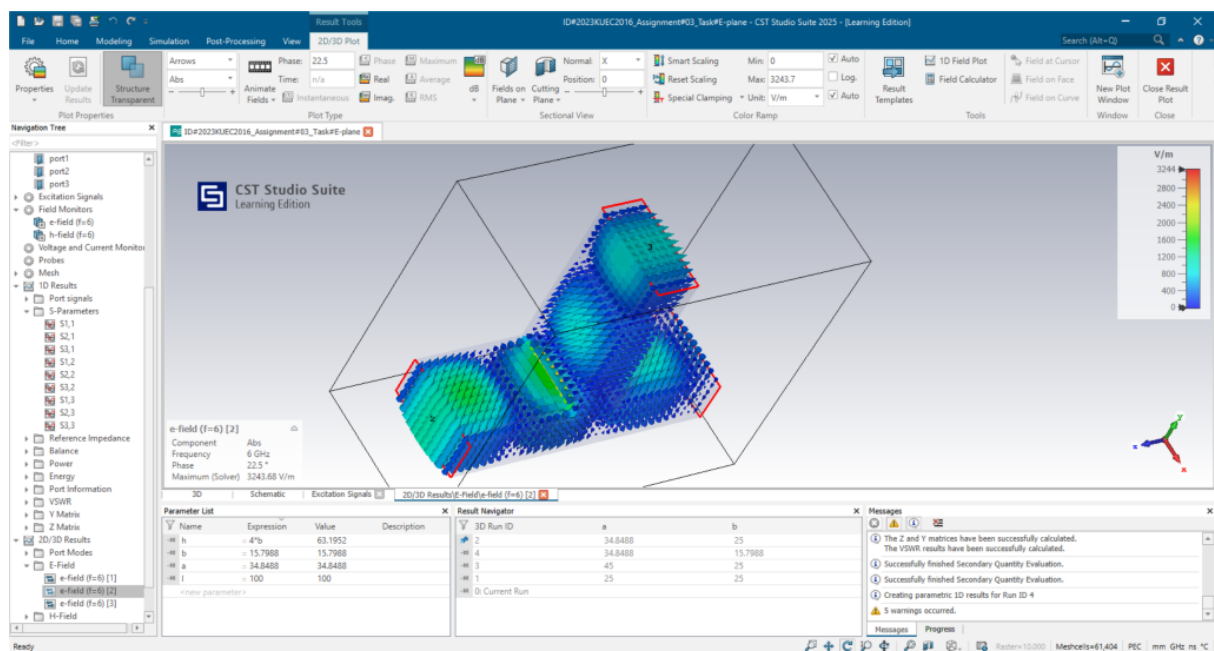


Fig 4.2 Electric field distribution for $f = 6$ GHz at port 2

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

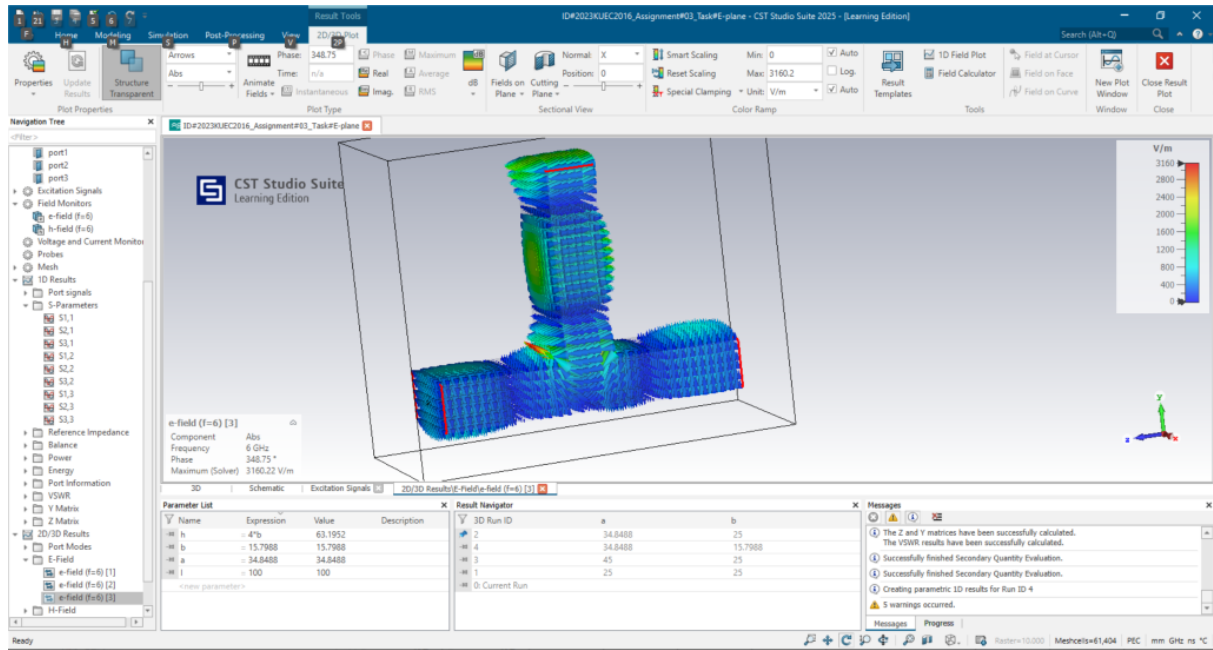


Fig 4.3 Electric field distribution for $f = 6$ GHz at port 3

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

Magnetic Field Distributions

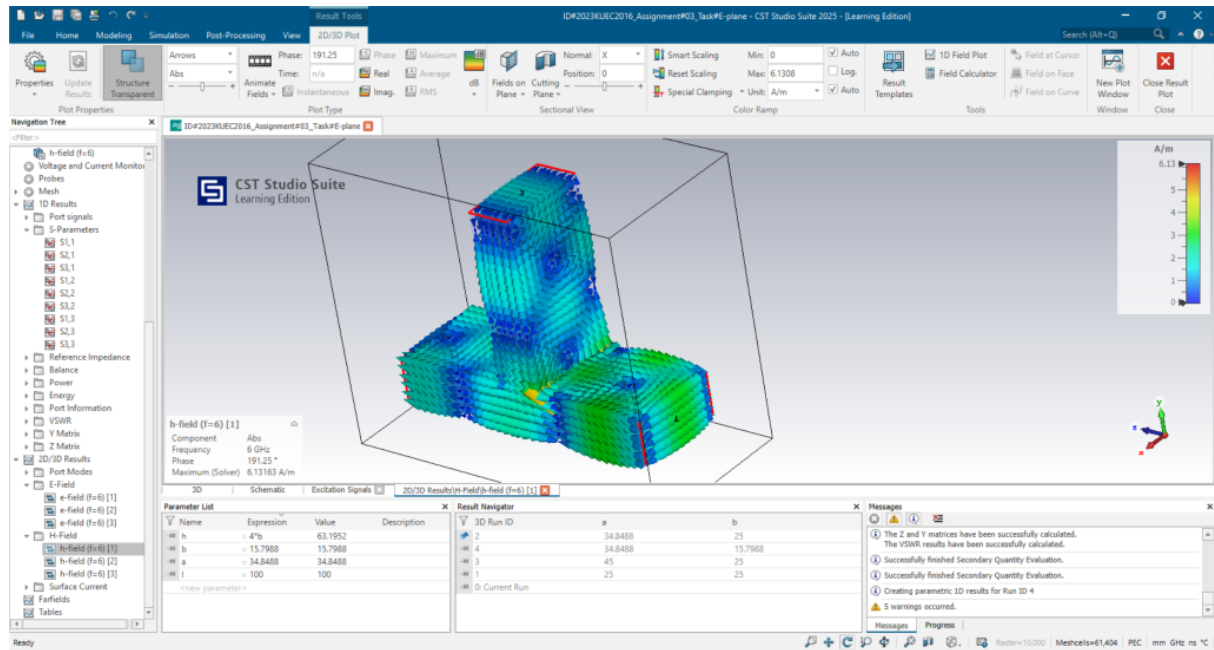


Fig 5.1 Magnetic field distribution for f = 6 GHz at port 1

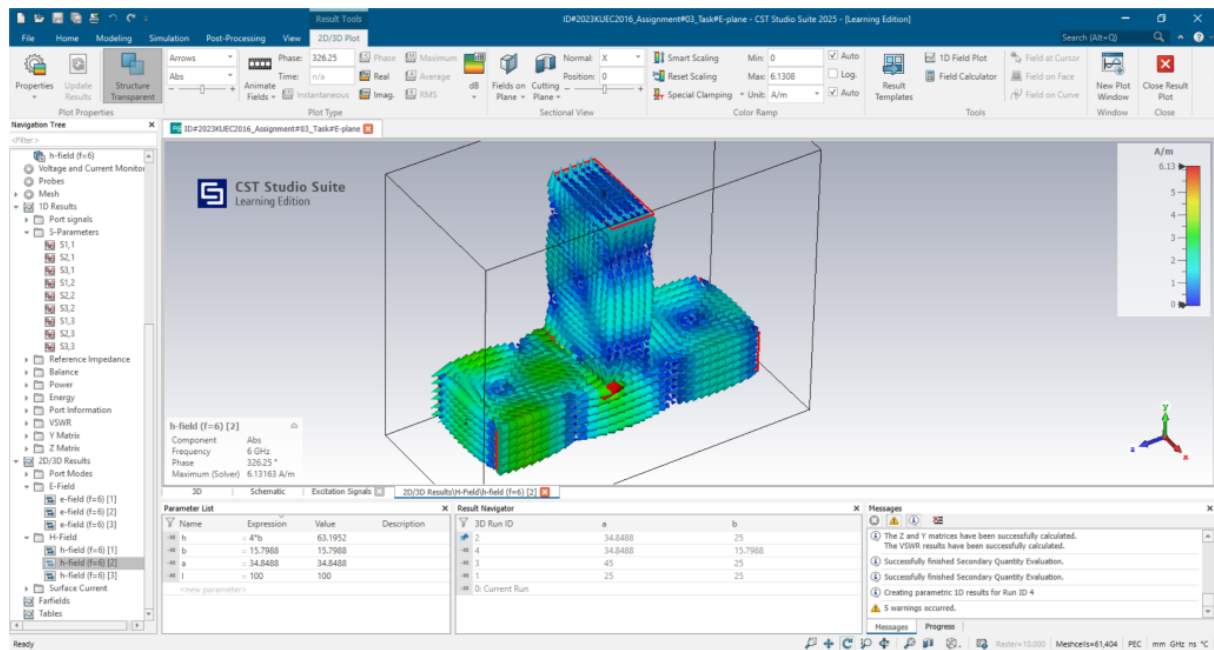


Fig 5.2 Magnetic field distribution for f = 6 GHz at port 2

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

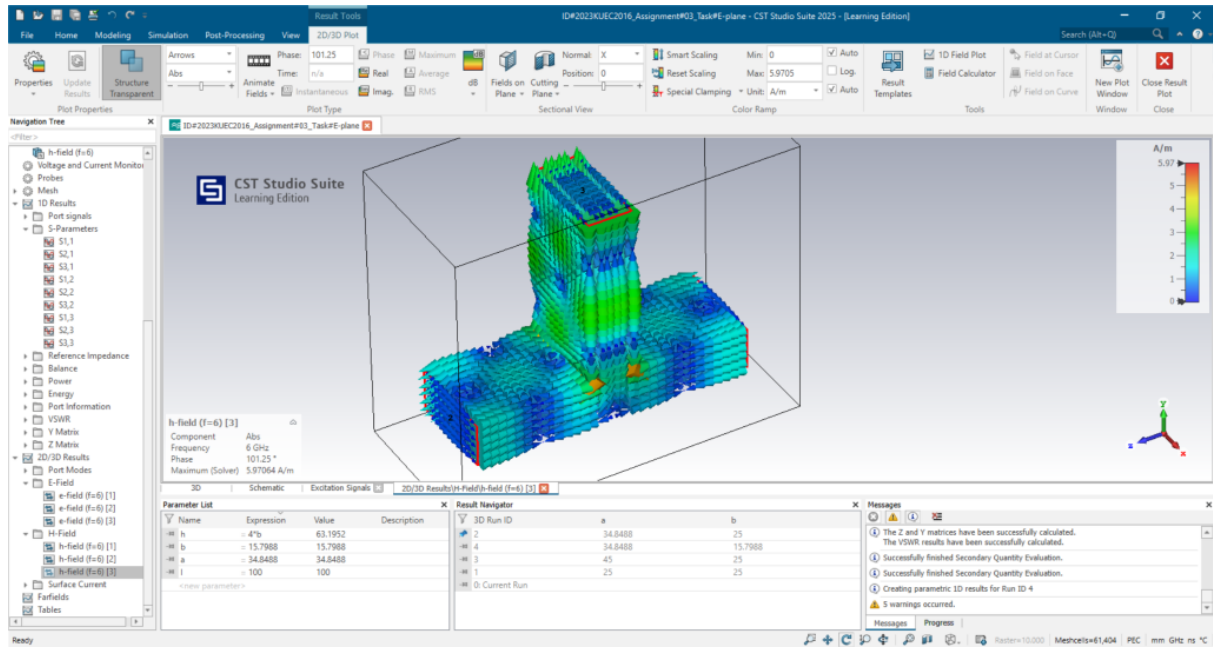


Fig 5.3 Magnetic field distribution for $f = 6$ GHz at port 3

H Plane Tee Junction

Scattering Parameters

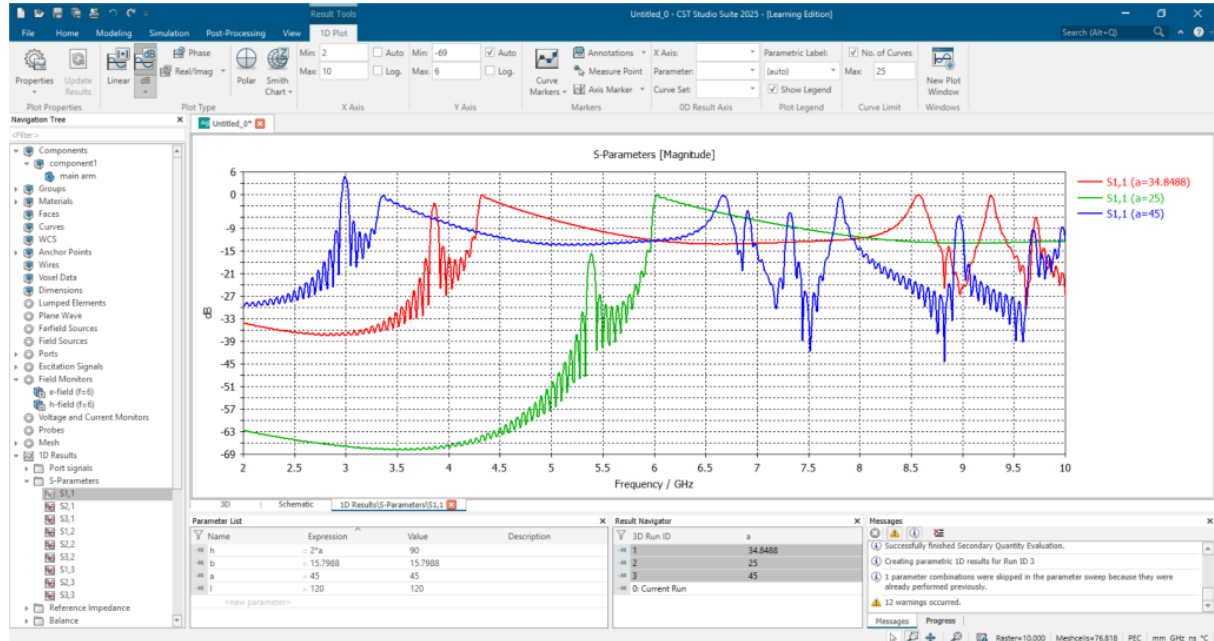


Fig 6.1 Scattering Parameter $S_{1,1}$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

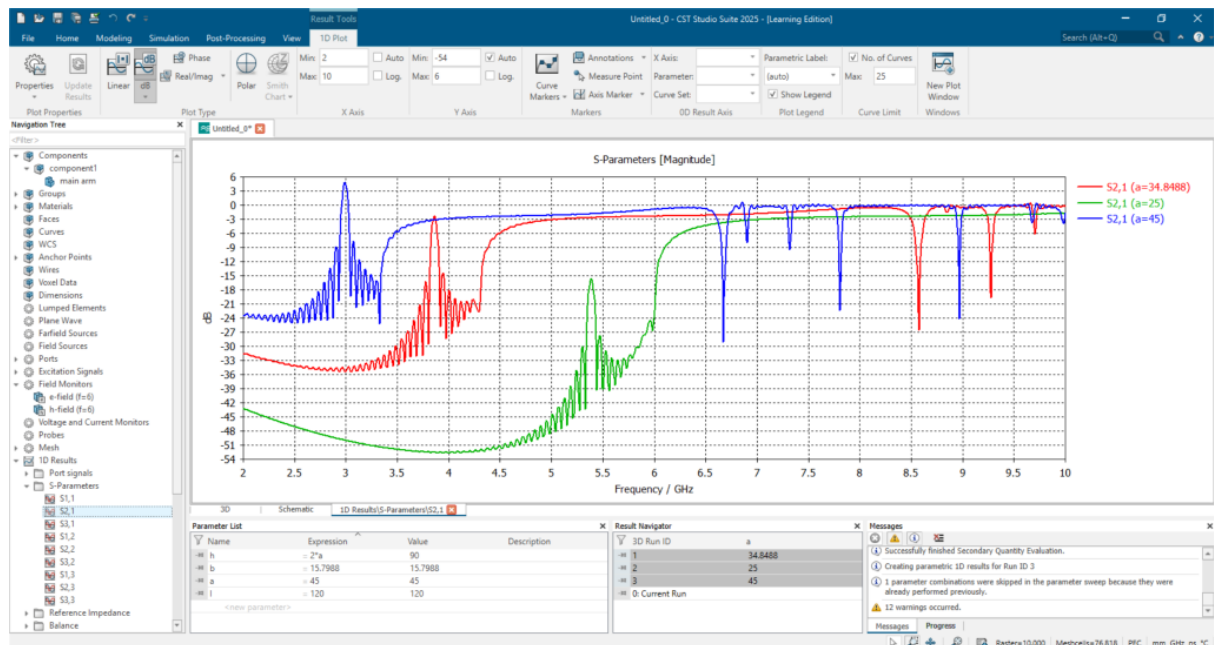


Fig 6.2 Scattering Parameter $S_{2,1}$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

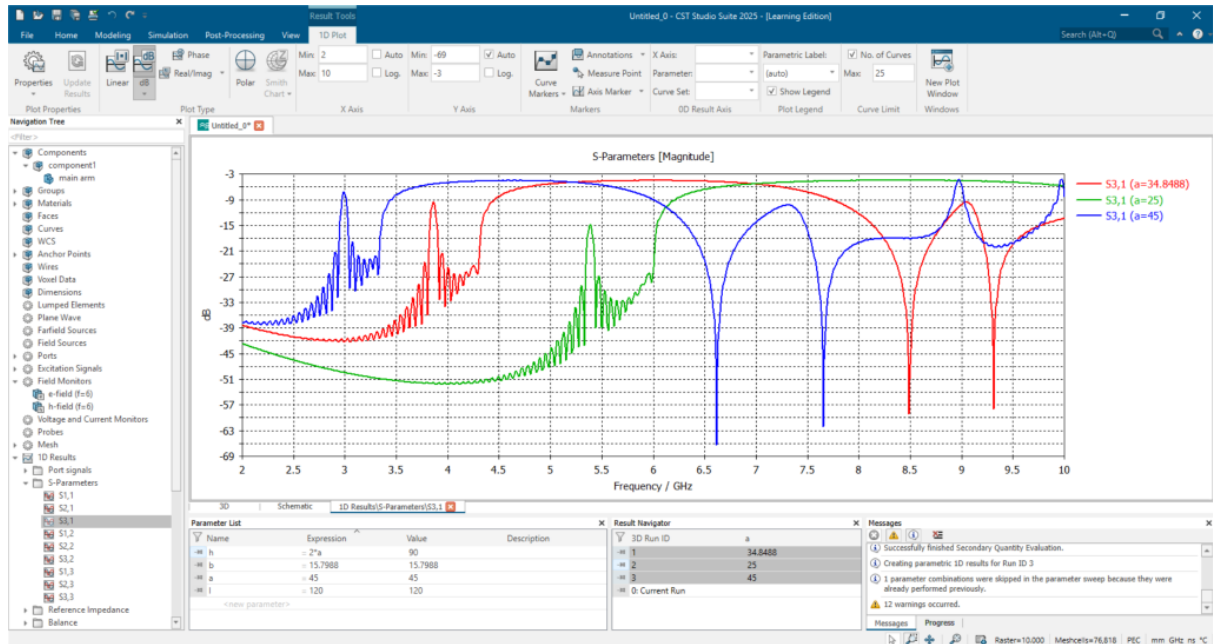


Fig 6.3 Scattering Parameter S(3,1) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

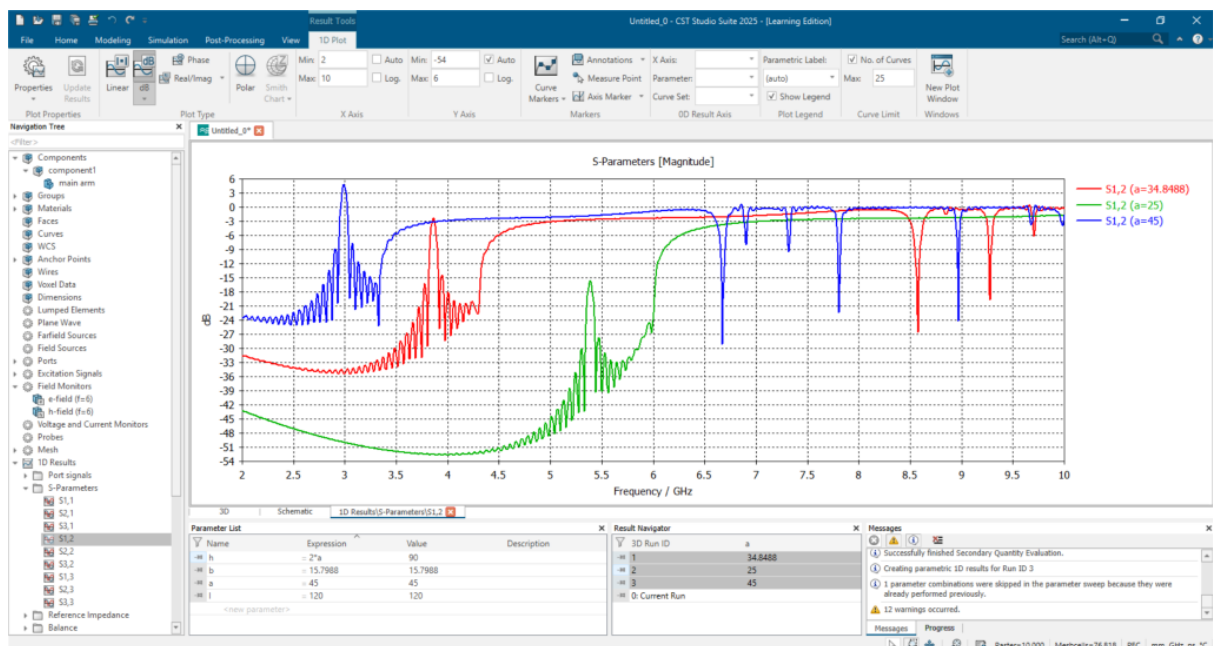


Fig 6.4 Scattering Parameter S(1,2) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

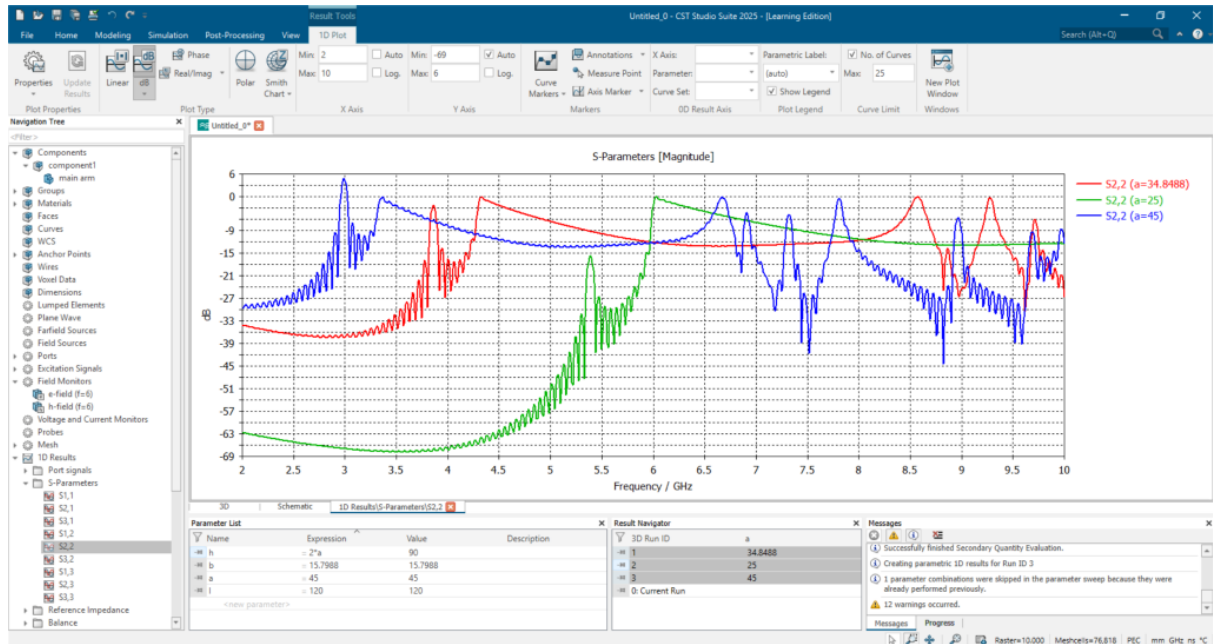


Fig 6.5 Scattering Parameter $S(2,2)$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

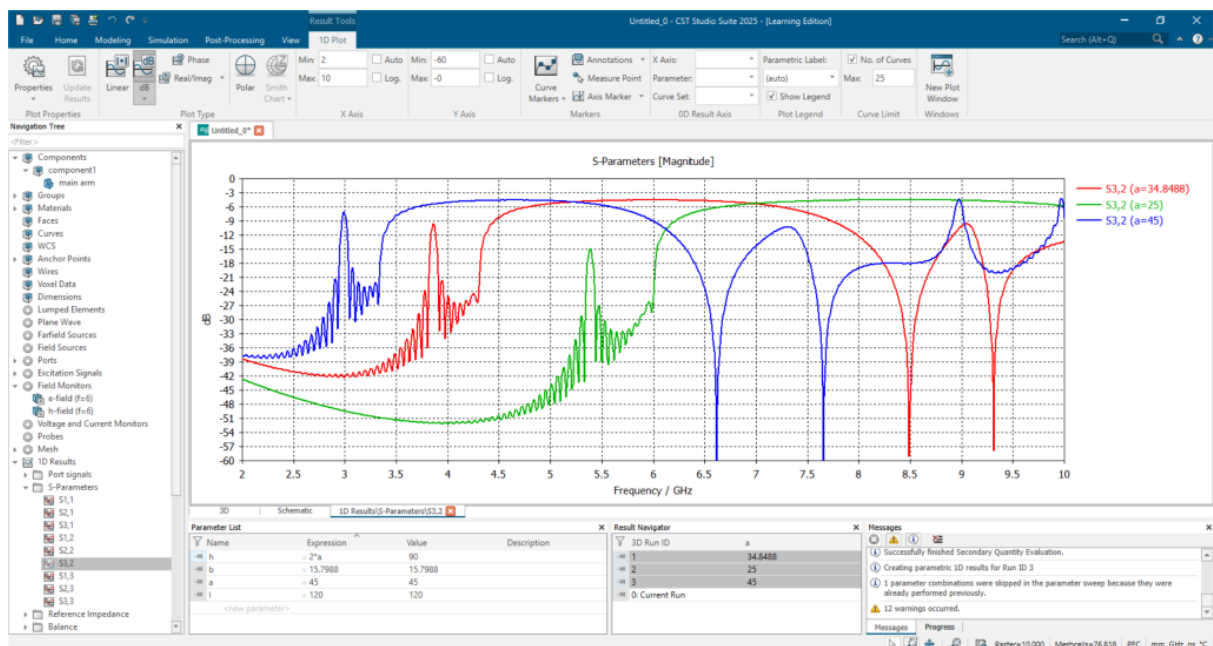


Fig 6.6 Scattering Parameter $S(3,2)$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

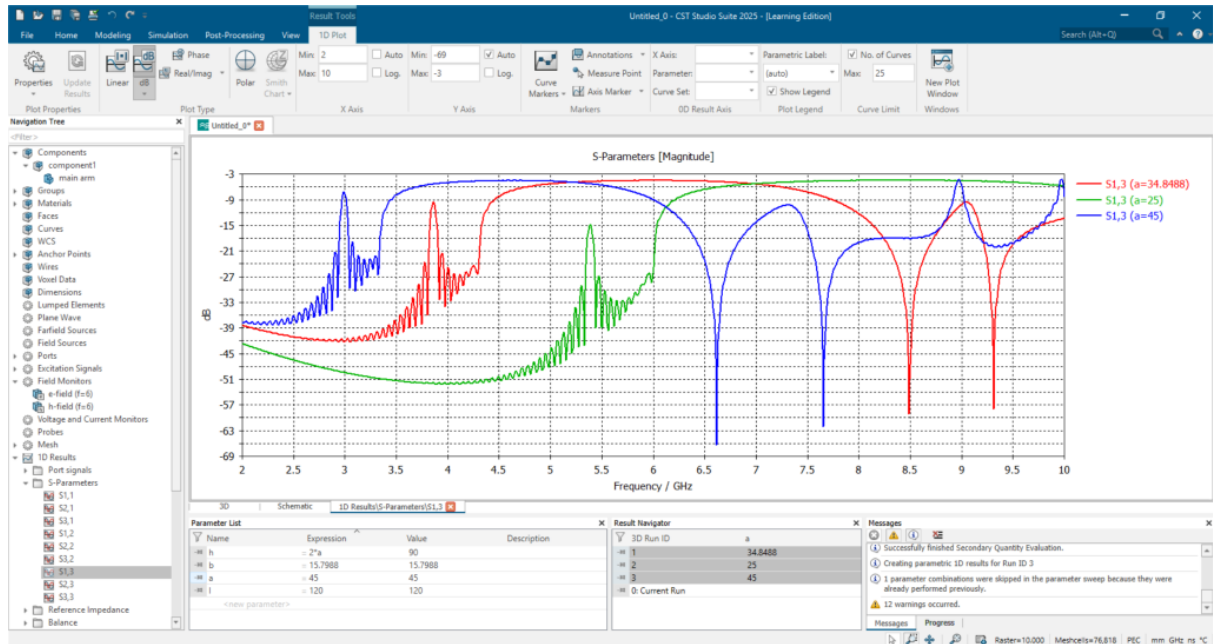


Fig 6.7 Scattering Parameter $S_{1,3}$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

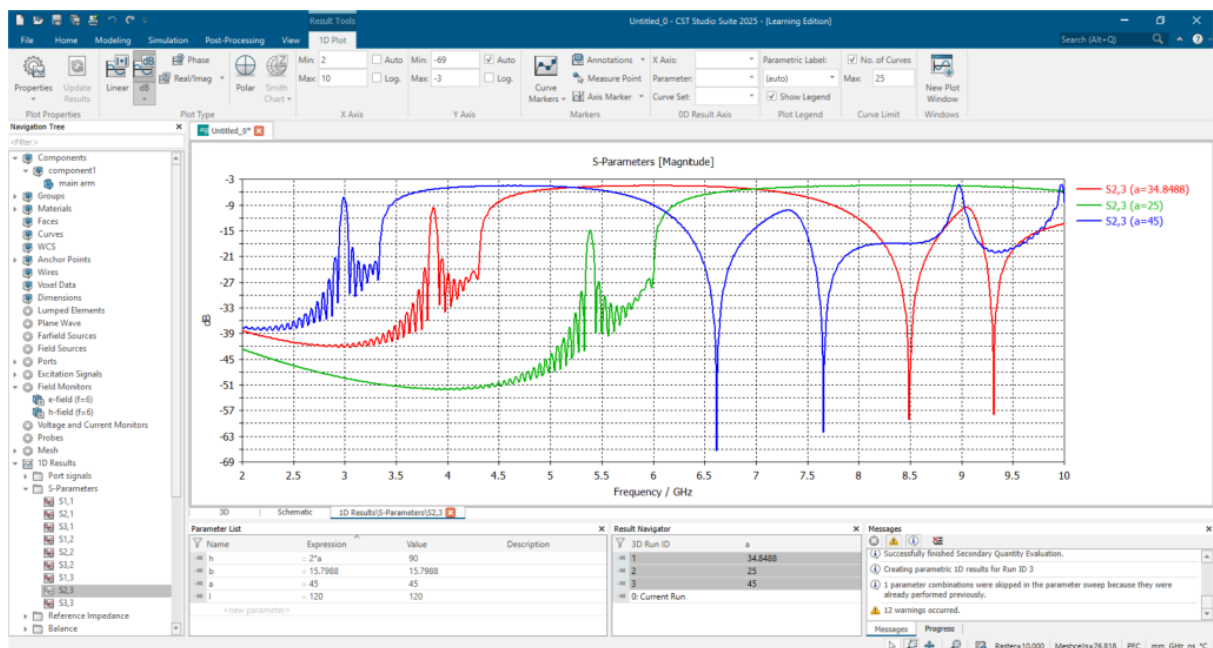


Fig 6.8 Scattering Parameter $S_{2,3}$ - Parameter sweep wider dimension ($a = 25, 34.8488$ and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

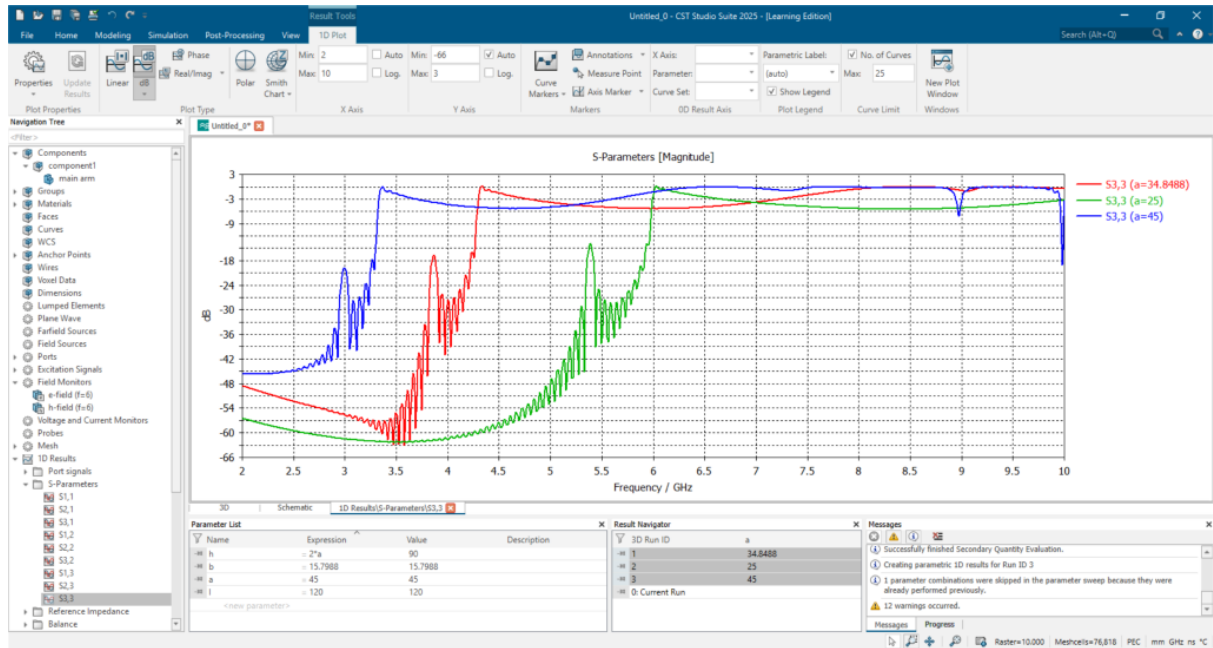


Fig 6.9 Scattering Parameter S(3,3) - Parameter sweep wider dimension (a = 25, 34.8488 and 45 mm)

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

Electric Field Distributions

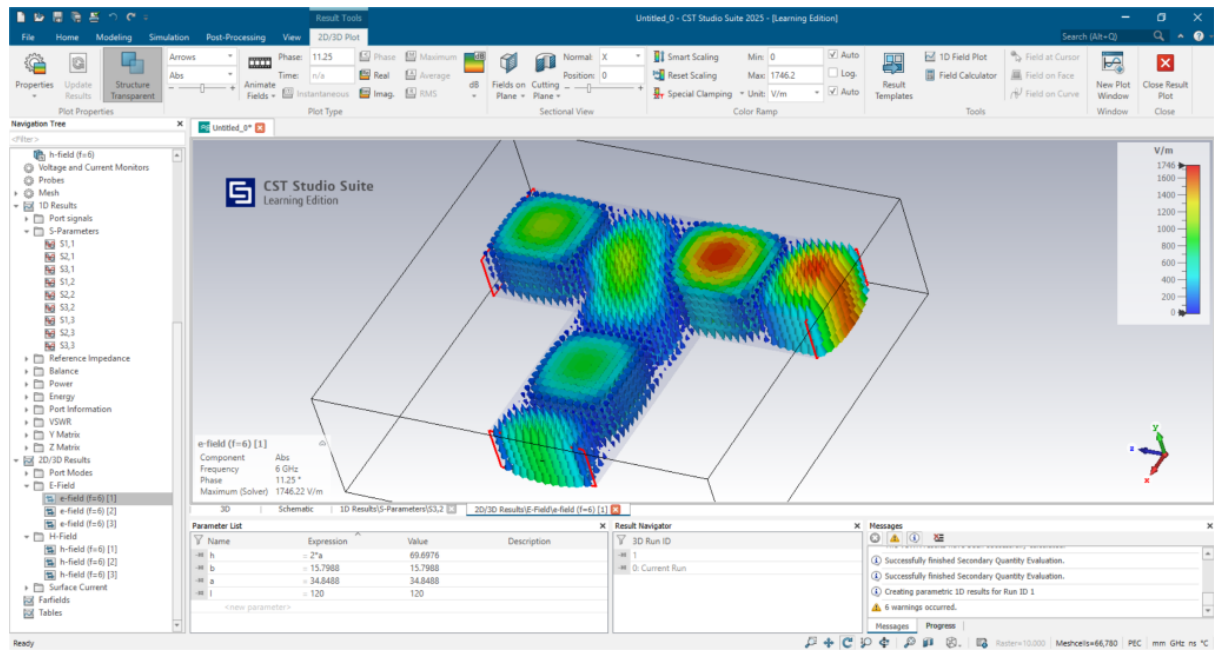


Fig 7.1 Electric field distribution for $f = 6$ GHz at port 1

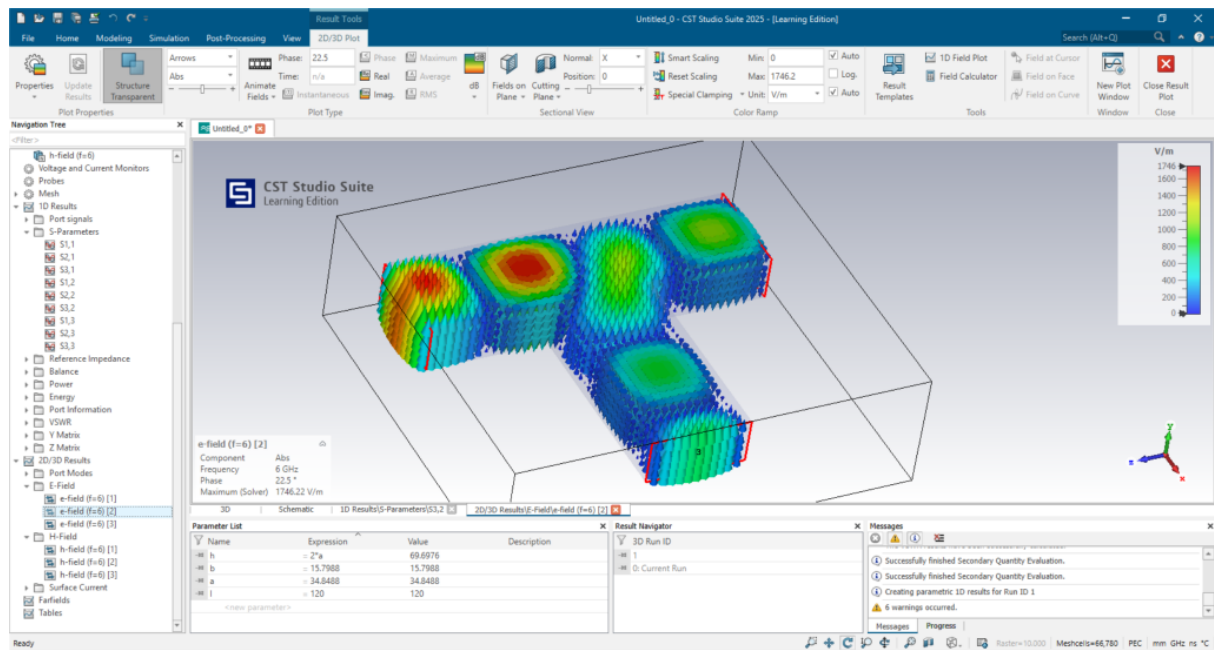


Fig 7.2 Electric field distribution for $f = 6$ GHz at port 2

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

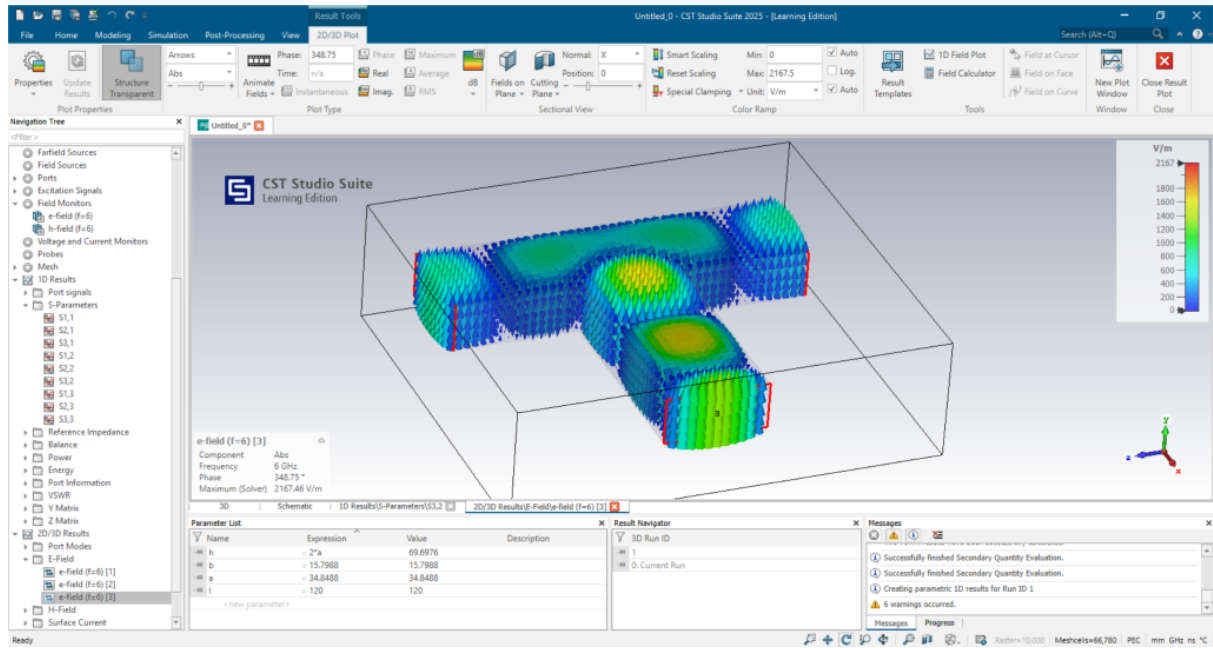


Fig 7.3 Electric field distribution for $f = 6$ GHz at port 3

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

Magnetic Field Distributions

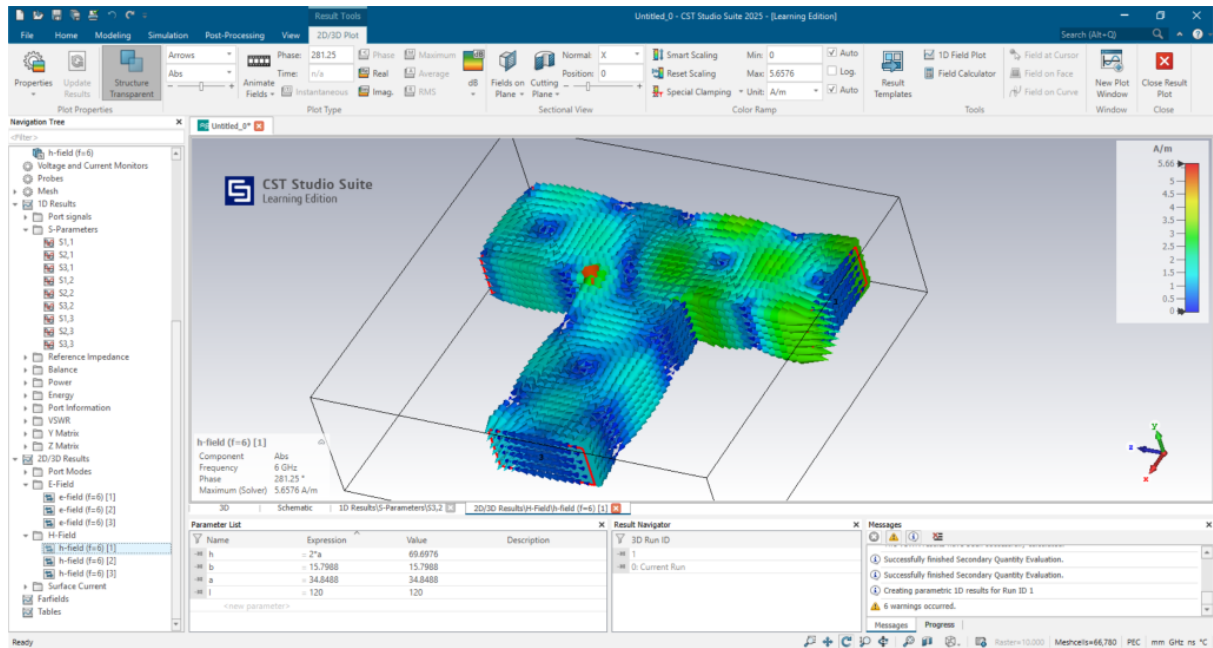


Fig 8.1 Magnetic field distribution for $f = 6$ GHz at port 1

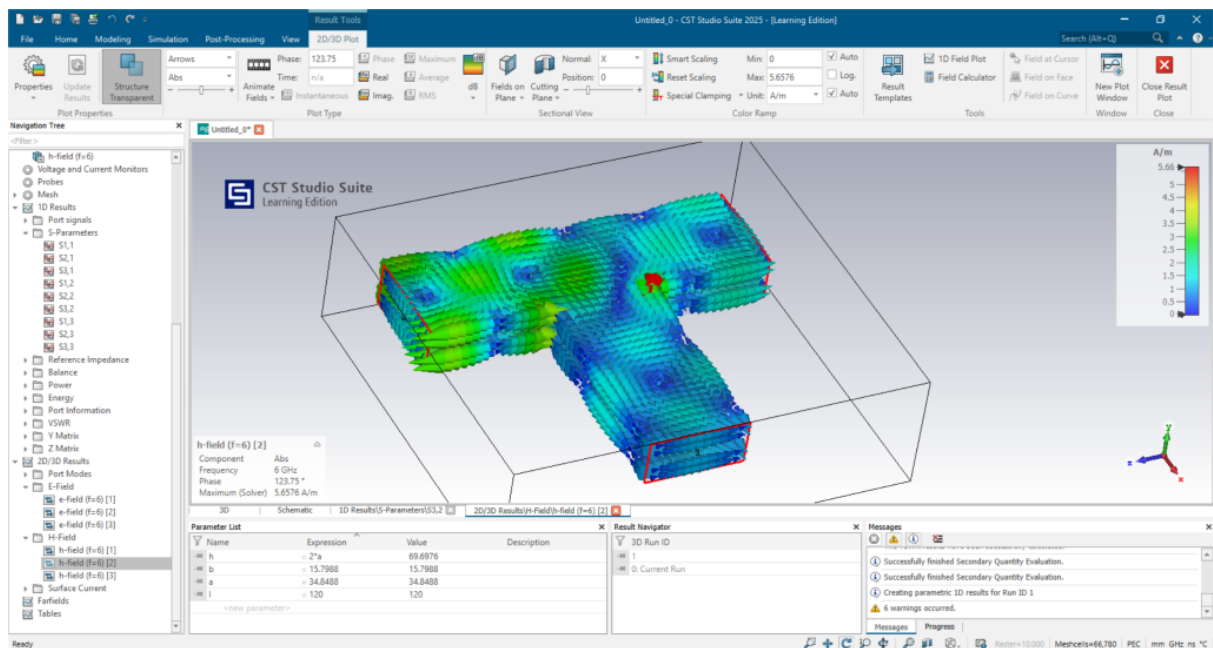


Fig 8.2 Magnetic field distribution for $f = 6$ GHz at port 2

Submitted by
Names: Sarthak Singh Tariyal
Roll Number.: 2023KUEC2016

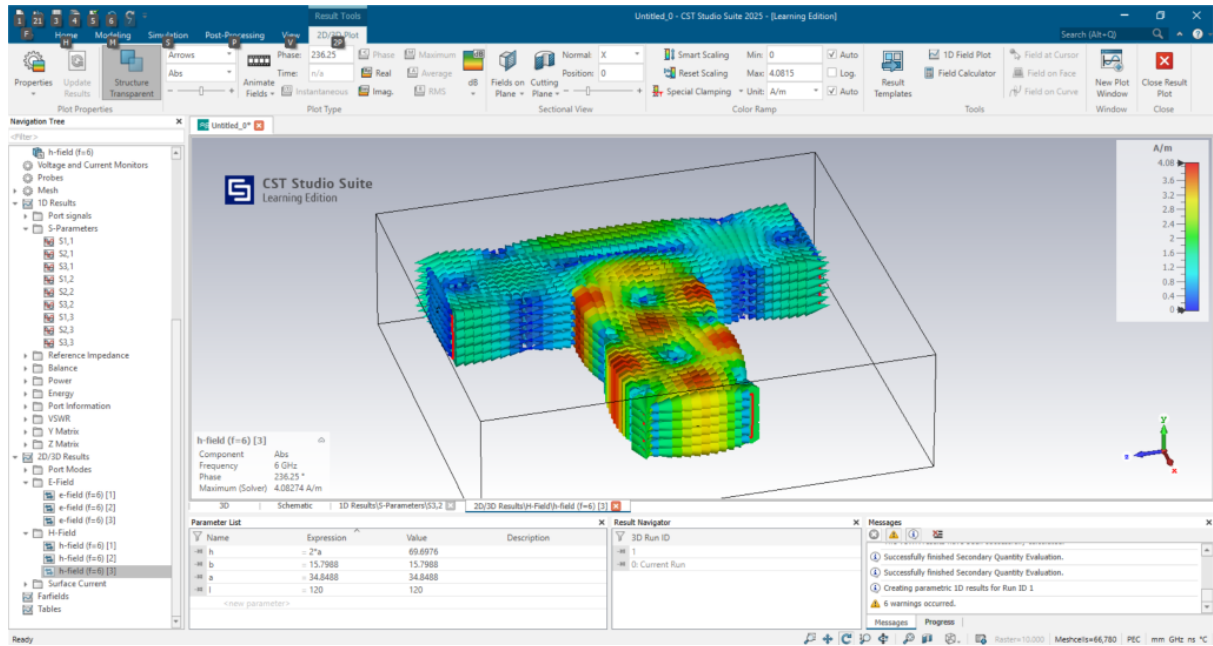


Fig 8.3 Magnetic field distribution for $f = 6$ GHz at port 3

Observations

S-Parameters and Cutoff Frequency Variations

The broadband frequency sweep from 2 GHz to 10 GHz provided critical insight into the relationship between the waveguide's physical width (a) and its propagation characteristics. As observed in the S-parameter plots, specifically Fig 3.2 (E-plane) and Fig 6.2 (H-plane), the "turn-on" frequency, where the transmission coefficient (S_{21}) rises sharply from negligible levels to near 0 dB, shifts significantly with the dimension a .

- For the 25 mm width (red curve), the cutoff frequency shifted upward to approximately 6 GHz, which is evident in Fig 3.2 where the signal is heavily attenuated below this mark.
- For the standard 34.8488 mm width (green curve), the cutoff remained at the theoretical 4.3 GHz, ensuring stable operation across the C-band.
- For the 45 mm width (blue curve), the cutoff dropped to approximately 3.3 GHz, allowing lower frequency propagation but introducing ripples suggestive of potential higher-order mode interference. Additionally, Fig 3.1 and Fig 6.1 (S_{11}) demonstrate that the return loss is minimized (indicating the best impedance match) only at the standard dimension, whereas the 25 mm and 45 mm variations show significantly higher reflection coefficients due to impedance mismatch.

E-Plane Tee

The operational behavior of the E-plane Tee as a series/subtractor junction was confirmed through both scattering parameters and field distributions. The S-parameter plots in Fig 3.2 (S_{21}) and Fig 3.3 (S_{31}) show that the power splitting magnitude is consistent across the operating band. Crucially, the phase-inversion characteristic was verified by the electric field distributions monitored at 6 GHz. In Fig 4.1, Fig 4.2, and Fig 4.3, the vector plots of the E-field illustrate that when power is fed into Port 3, the fields coupling into the main arm (Ports 1 and 2) are 180° out of phase. This anti-symmetric field pattern confirms the theoretical S-matrix property $S_{13} = -S_{23}$.

H-Plane Tee

The H-plane Tee's function as a shunt/adder junction was similarly validated. The S-parameter plots in Fig 6.2 (S_{21}) and Fig 6.3 (S_{31}) indicate symmetric power division with equal magnitudes. The "in-phase" property was visually confirmed using the electric field monitors at 6 GHz shown in Fig 7.1, Fig 7.2, and Fig 7.3. Unlike the E-plane tee, the field vectors in the H-plane junction branch out symmetrically from the side arm into the main arm, maintaining the same field orientation for both outputs. This confirms the S-matrix property $S_{13} = S_{23}$, identifying it as a coherent power divider.

Submitted by

Names: Sarthak Singh Tariyal

Roll Number.: 2023KUEC2016

Conclusions:

The design and simulation of E-plane and H-plane Tee junctions using the WR-137 waveguide standard were successfully completed, verifying the fundamental scattering properties of microwave three-port networks. The observed S-parameters aligned with theoretical expectations: the E-plane Tee demonstrated characteristics of a phase-inverted difference junction, while the H-plane Tee functioned as an in-phase sum junction. These behaviors were definitively corroborated by the 3D electric and magnetic field distributions at 6 GHz, which visually depicted the anti-symmetric coupling for the E-plane and symmetric coupling for the H-plane.

Furthermore, the parametric sweep of the wider dimension (a) highlighted the critical dependence of the cutoff frequency (f_c) on waveguide geometry. It was observed that decreasing the width to 25 mm raised the cutoff frequency to approximately 6 GHz, effectively rendering the waveguide evanescent for the lower half of the C-band. Conversely, increasing the width to 45 mm lowered the cutoff to 3.3 GHz, allowing propagation but compromising impedance matching, as evidenced by the degraded return loss (S_{11}) in the deviation cases. This confirms that the standard dimension of 34.8488 mm is essential for optimizing bandwidth, mode stability, and impedance matching in C-band applications.